

Manoj Kumar Selvaraj

Sr. ASSET INTEGRITY & INSPECTION ENGINEER

13+yrs-Mechanical Integrity | RBI IOW | NDT | FFS| DM's | Turn around | RCA | Inspection/TA Optimization| Material & Corrosion Management | Budget Handling| Repair & Re- Rating | AIMS | Technology Adaptation.



escmanojkumar@gmail.com
+966534360536, +91 9940558335

Residence: KSA, Home Country: India
S. Manoj Kumar

07 July 1989

<https://www.linkedin.com/in/manoj-integrity-engineer/> @dot7technicals

CURRENT POSITION & EMPLOYEE

SPIE Oil & Gas Services (For JIGPC)
Inspection Engineer & Interim Sr. Integrity Engineer

01/2023- till date Jizan, KSA (world's largest IGCC - Gasification)

CORE COMPETENCE.

- INSPECTION & INTEGRITY MANAGEMENT:** On/Off-stream inspections, Technical/Operational Integrity, MOC execution Damage Mechanism-specific strategies, SAP-integrated workflows, Corrective/Preventive/ Predictive Maintenance
- STATIC EQUIPMENT EXPERTISE:** Pressure vessels, PRDs, exchangers, columns, conductor casing, tanks,5-cone gasification burners, refractory systems
- RBI & FFS PROFICIENCY:** API 580/581-based qualitative RBI, RCA studies, Fitness-for-Service assessments - Fire, Blister.
- TURNAROUND & RELIABILITY:** Led 15+ turnarounds, R-stamp repairs, downgraded system strategies, Inspection deferral, TA scheduling.
- CORROSION & MATERIAL ENGINEERING:** Corrosion mitigation, material selection, re-rating evaluations, dead-leg elimination, CCD & IOW development, RCA study.
- WELDING & NDT:** PWPS, PQR & WPS Generation. Weld defect troubleshooting, WPS/PQR audits, ITP documentation Skilled in C-NDT/A-NDT techniques, UT & MFL-intelligent pigging system, PMI Analyzer, Scorpion B-Scan - Remote Access, Magnetic Flux Leakage. Vacuum Box Testing, Borescope. Interpretation Skill: MFL, RFET, Eddy Current, IRT, Leak detection IRT.
- INNOVATION/NEW TECHNOLOGIES & OPTIMIZATION:** CML/Turnaround/Inspection Strategy optimization Implementation of smart inspections, technical alerts, and emerging reliability tools.
- MANAGEMENT & BUDGET HANDLING:** Capex & Opex budget handling, Audit documentation
- COMPUTER & SUPPLEMENTARY SKILLS:** Maximo, SAP, E-Passport, Intergraph Free view, Auto CAD. AI Tools: Chat GPT/Copilot, Power BI. MS office suite.

AWARDS

Employee of the year -2018 at Inspecta International
HSE Man of the year -2019 at Inspecta International
Integrity Excellence in Defect Management - 2023 at JIGPC
Custodian of Plant Integrity- 2024 at JIGPC

EDUCATION

Bachelor Degree in Mechanical Engineering. - Anna University
Professional Engineer: PE No: PE7007472 from The Institution of Engineers (India)

AUTHORED

Inspection Guidelines for heat Exchanger Book: ISBN 978-93-6249-409-2

TECHNICAL CERTIFICATES

API 510 (Pressure Vessel Inspector): Certificate No - 79056

API 653(Storage Tank Inspector) : Certificate No - 72170

API 580(Aboveground Storage Tank Inspector) :
Certification No - 105436

API 570 (Piping Inspector): Certificate No - 64667

API 571 (Corrosion & Materials Professional); Certificate
No - 109457

API 936 (Refractory Inspection) Certificate No-
113505

CSWIP 3.2.1(Sr. Welding Inspector): Certificate No -
676783

ASNT Level-III (DPT, MPT, RT & UT): Certificate No -
316598

NACE CIP Level I (Coating Inspector): Certificate No -
11162873

SERVICE'S EXPOSURE:

Gasification - Vacuum Residue to Syngas, Acid Gas Removal, Hydrogen Recovery Unit -Pressure Swing Adsorption, Sulphur Recovery Unit, Power Generation - with Syngas, Oxygen Services, Utilities. Steam Generation With HRSG, Upstream Industries: At Onshore & offshore - Crude and Gas Extraction.

ENGINEERING CALCULATION EXPERTISE:

Minimum required thickness of piping, pressure vessels & exchanger tubes.

Bolt torque values.

Mechanical Temporary repair clamp design calculation.

Storage tank bottom plate thickness, fill height.

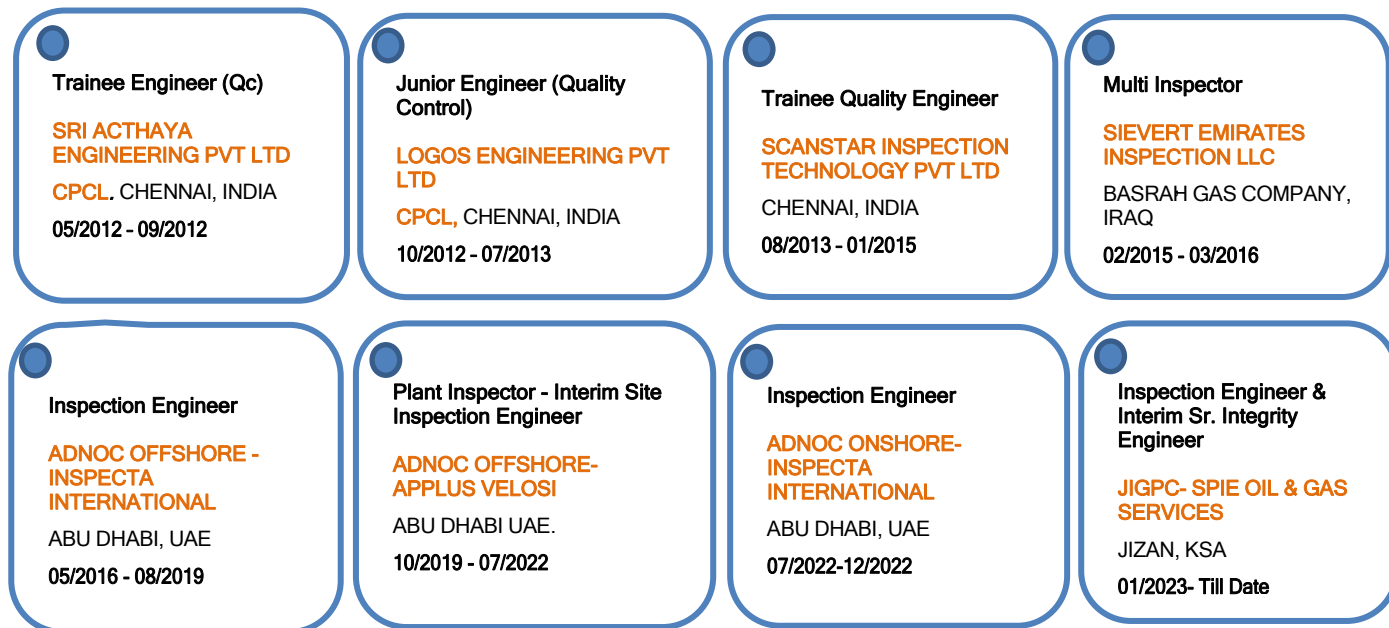
Maximum allowable operating pressures.

Corrosion rate, remaining life calculations. And ability to review various engineering calculations.

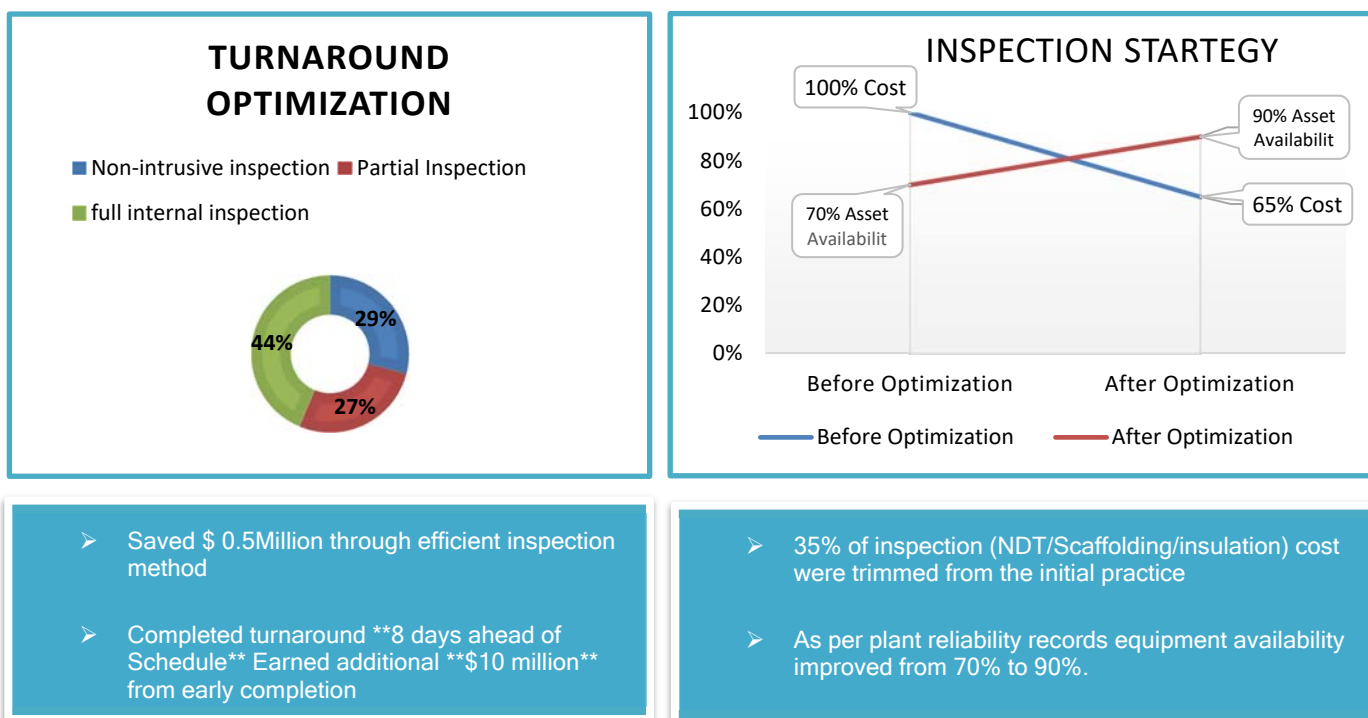
EXPOSURE ON:

Fault tree analysis, fish bone methodology, Swiss cheese model, 5whys, HAZOP, IOW, Bow tie model, SIMOPS, ECA, FARCAS, MTBF, RCM, RAMS, HMB, Cause & Effect Diagram, corrosion loop, inventory loop, PFD, MSD, BOD

EXPERIENCE HISTORY (POSITION & EMPLOYEE).



MAJOR FINANCIAL ACHIEVEMENTS:



MAJOR TECHNICAL ACOMPLISHMENTS

1. Turnaround Optimization

- **S/T:** Faced a critical shutdown window with over 300 equipment requiring inspection.
- **A:** Applied RBI to prioritize scopes; adopted non-intrusive/partial intrusive methods and deferred low-risk items.
- **R:** Completed turnaround 8 days early and achieved \$10.5M in total cost/time savings.

2. CML Rationalization

- **S/T:** Excessive inspection workload from 13450+ CMLs.
- **A:** Rationalized CMLs by 40% through corrosion loop review and criticality assessment.
- **R:** Reduced man-hours by 35% and improved data accuracy.

3. Fitness-for-Service Assessment

- **S/T:** Multi Hearth furnace, screw conveyer, oxygen preheat condensate affected by fire incident.
- **A:** Conducted Level 1 FFS as per API 579 and optimized inspection intervals.
- **R:** Safely extended service life by 5 years, saving over \$12M in replacement costs.

4. RBI Revalidation with Corrosion Loop Optimization, Circuitization Enhancement & CCD Update

- **S/T:** Existing RBI database was outdated; lacked integration with recent inspection findings and process changes, and CCDs were not aligned with actual corrosion behavior in field.
- **A:** Led a full RBI revalidation campaign—updated damage mechanism reviews, refined corrosion loops based on actual inspection data and process parameters, and enhanced circuitization by integrating changes from MOC and operational feedback. Revised and standardized Corrosion Control Documents (CCDs) to reflect updated degradation rates, mitigation strategies, and IOW compliance.
- **R:** Improved risk accuracy and inspection coverage; reduced overall inspection backlog by 38%, prioritized 20% of circuits for extended intervals based on updated risk ranks, and improved audit readiness and regulatory compliance across the plant.

5. Failure Investigation Leadership

- **S/T:** Unplanned failure of a flare acid gas system, gasification burner failure, light slope Vaporizer & various occurred.
- **A:** Led RCA using, process trends, C&I narratives, lab analysis and failure morphology.
- **R:** Identified the root cause and revised material standards/implemented best practices to prevent recurrence.

6. Crack Detection and Fitness-for-Service Assessment

- **S/T:** Detected axial surface cracks on a high-pressure steam line during phased array ultrasonic testing (PAUT) as part of on-stream inspection.

- **A:** Performed API 579 Level 3 Fitness-for-Service (FFS) assessment including fracture mechanics evaluation calculated critical flaw size, and verified with actual operating stresses and material toughness.
- **R:** Demonstrated that crack was non-critical under current conditions; avoided emergency replacement. Recommended weld repair during next turnaround and safely extended service life by **18 months**, saving ~\$500K in immediate repair and shutdown cost.

7. AIMS Implementation

- **S/T:** Asset integrity data was fragmented across teams.
- **A:** Deployed a full Asset Integrity Management System aligned with API 580/581.
- **R:** Achieved 40% fewer failures and improved regulatory compliance.

8. Inspection Strategy Consolidation

- **S/T:** Redundant vendor scopes increased cost.
- **A:** Streamlined strategies using integrity workshops and risk-ranking tools.
- **R:** Reduced inspection costs by 25% annually.

9. On-Stream Inspection Execution

- **S/T:** Limited time for internal inspection.
- **A:** Deployed on-stream UT, EMAT, Intelligent pigging, guided wave tools, profile RT and various onstream NDT techniques.
- **R:** Extended inspection coverage and reduced shutdown dependencies.

10. Corrosion Control Program

- **S/T:** Corrosion rates exceeded limits in sour services.
- **A:** Implemented chemical dosing, and online monitoring.
- **R:** Reduced corrosion rates by 50% and prolonged piping integrity.

11. IOW Development

- **S/T:** Lack of defined process envelopes led to excursions.
- **A:** Enforced tracking Integrity Operating Windows (IOW) across critical units.
- **R:** Prevented 2 major integrity incidents and improved process control.

12. ASME-Stamped Pressure Vessel Repair & Integrity Restoration

- **S/T:** Discovered localized internal corrosion and pitting in an ASME Section VIII-stamped pressure vessel during internal inspection. Vessel was critical for continuous plant operations.
- **A:** Led detailed thickness evaluation, conducted API 579 Level 2 Fitness-for-Service, and coordinated a coded repair plan including shell patching and weld overlay. Managed repair contractor qualified under NBIC R-Stamp requirements.

- **R:** Successfully restored vessel integrity, retained ASME Code Stamp without de-rating, and avoided full vessel replacement—saving over \$20 million and 21 days of production loss.

13. MOC Governance

- **S/T:** Engineering changes lacked integrity validation.
- **A:** Led technical MOC board for corrosion, metallurgy, and mechanical changes.
- **R:** Zero unauthorized changes and full traceability maintained.

14. PRD Compliance and Optimization

- **S/T:** Relief devices required recertification backlog clearance.
- **A:** Created PRD integrity plan and initiated in-house testing.
- **R:** Cleared backlog; reduced outsourcing cost by 40%.

15. Valve Integrity Framework

- **S/T:** High valve failure rate in critical systems.
- **A:** Established leakage criteria, test frequencies, and repair matrix.
- **R:** Improved valve reliability and reduced repeat failures by 25%.

16. Budget & Resource Efficiency

- **S/T:** Rising inspection and manpower costs.
- **A:** Introduced inspection bundling and vendor optimization.
- **R:** Cut inspection budget by 15% while maintaining performance KPIs.

17. KPI-Driven Performance Management

- **S/T:** Lack of visibility into integrity team performance.
- **A:** Developed KPIs (compliance %, failures, backlog aging).
- **R:** Enabled better planning and reduced overdue inspections by 30%.

18. Root Cause Analysis (RCA) & Implementation

- **S/T:** Recurring corrosion failures in same locations.
- **A:** Standardized RCA using 5-Why and Fishbone.
- **R:** Reduced repeat failures by 60% and drove proactive actions.

19. Process Logic & Trend Analysis

- **S/T:** Difficult to trace hidden process faults linked to failures.
- **A:** Used DCS trends, PFDs, and logic diagrams to identify causes.
- **R:** Prevented 4 critical events by acting on abnormal process patterns.

20. Technical Documentation & Data Management

- **S/T:** Incomplete integrity records caused delays during audits.
- **A:** Developed centralized database for inspection reports, repair history, and manuals.
- **R:** Cut audit response time by 80% and improved knowledge retention.

21. Welding Program

- **S/T:** Needed qualified procedures for Hastalloy, Monel, WCB, Inconel, SS316,321,347,304 & dissimilar joints.
- **A:** Developed pWPS, tested PQR and derived WPS as per ASME IX and construction code.
- **R:** Prevented potential weld failures; ensured long-term reliability in high-service areas.

22. Non-Welding Online Repairs in Sour Service

- **S/T:** Sour service leak risked shutdown.
- **A:** Applied online composite wrap and ASME PCC-2 clamp-based repair.
- **R:** Restored functionality without shutdown, saving 48 hours of downtime.

